Catalog No. C12118A

DADCO®

Micro Nitrogen Gas Springs

Micro Series

PED
97/23/EC
COMPLIANT

Additional Threaded
Body Models Included!



Ideal for Coil Spring Replacement

Introduction

DADCO produces top quality products at competitive prices and provides a superior level of customer service. Founded in 1958, DADCO is the highest volume producer of gas springs for press tools. DADCO's products are widely approved and used in global operations for many industries including metal stamping, automotive and plastic injection molding.

Advanced Technology

DADCO's revolutionary Micro Nitrogen Gas Springs provide unparalleled versatility in industrial tools. The patented design offers unmatched performance in high quality dies, molds and machines.

Nitrogen Gas Springs vs. Coil Springs

DADCO's Micro Nitrogen Gas Springs easily replace conventional coil springs. Micro Springs deliver more force in less space than coil springs and one Micro Spring can provide the force of several heavy-duty coil springs. See page 3 for Coil Spring conversion information.

Range of Micro Sizes

DADCO's Micro Nitrogen Gas Springs are available in eight models.

Model	Diameter	Maximum Force on Contact	Threaded Body
Micro 45®	12 mm *(.472˝)*	112 lb. *(50 daN)*	*M16 x 1.5*
			M16 x 2
			5/8˝-11
Micro 70®	15 mm *(.591˝)*	154 lb. *(68 daN)*	*N/A*
Micro 90®	19 mm *(.748˝)*	200 lb. *(89 daN)*	*M24 x 1.5*
			1˝-8
Micro 180®	25 mm *(.984˝)*	450 lb. *(200 daN)*	*N/A*
Micro 250®	32 mm *(1.260˝)*	701 lb. *(313 daN)*	*N/A*
E.16	12 mm *(.472˝)*	95 lb. *(42 daN)*	*M16 x 1.5*
SL.16	14 mm *(.551˝)*	114 lb. *(51 daN)*	*M16 x 1.5*
E.24	21.5 mm *(.846˝)*	381 lb. *(170 daN)*	*M24 x 1.5*

The **Micro 45®**, **Micro 70®**, **Micro 90®**, **Micro 180®** and **Micro 250®** gas springs are color-coded for easy identification of force rating and are shipped ready to install. No additional equipment or previous experience with nitrogen is required.

DADCO offers adjustable force models that can be customized to meet individual force requirements. The adjustable model may be set to the desired pressure at the factory or in the field with appropriate charging equipment. See page 18 for more information on charging Micro Springs.

Threaded Body

DADCO's Threaded Body models are ideal part ejectors, replacing conventional coil spring stock lifters. The E.16, SL.16 and E.24 are designed with domed rods to further facilitate part removal during operation. DADCO provides a variety of hex tools, shown on page 17, that allow for easy installation and removal of the Threaded Body models.

Cost Effective

DADCO Micro Springs are inexpensive, easy to install, and provide a cost-effective solution to downtime problems associated with other springs.

No Preloading Required

DADCO Micro Springs deliver full rated force on contact with no preloading required. Occasionally, slight preloading is recommended, especially for stroke lengths from 150-200 mm, to prevent full spring travel where material thickness varies and parts can stick (i.e. stripper applications).

Rod Wiper Fights Draw Compound

DADCO's Duralene® Rod Wiper excludes most draw die compounds. For applications where an aggressive draw die compound is used contact DADCO for alternative wiper options.

Guaranteed Long Life



In factory testing and field experience, the service life of DADCO's Micro Springs consistently exceeds one million strokes. This is supported by DADCO's written One Year/One Million Stroke ***Gold Guarantee***. Contact DADCO or your representative for more information.

CAD Templates On-line

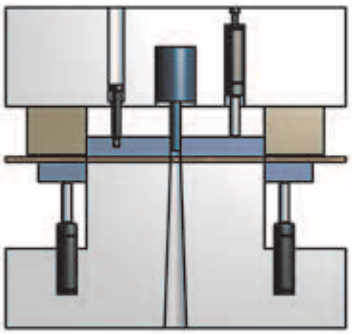
DADCO's entire product line is available on-line in solid models and 2D CAD formats. For more information, visit our website, www.dadco.net, or contact DADCO.

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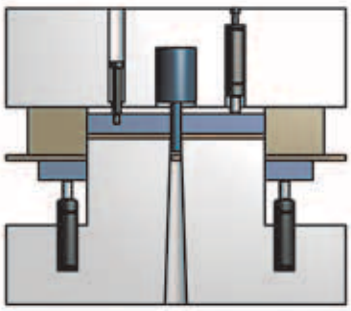
Application Examples

Blanking

INNER PRESSURE PAD AND PUNCH STRIPPER CLAMPS STOCK

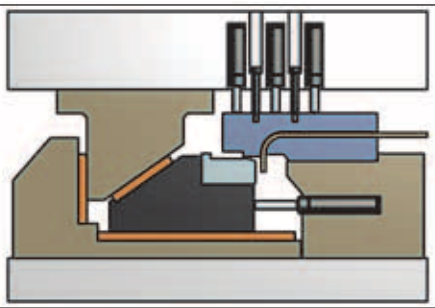


BLANKED PART SHOWN WITH BOTH PUNCH AND TRIM STRIPPERS COMPRESSED

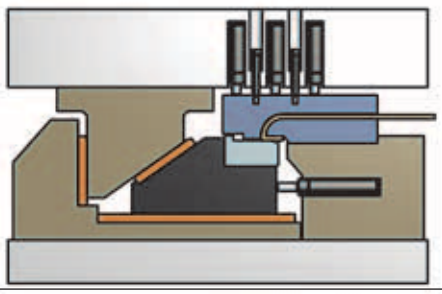


Bend and Flange

LEADING SPRING PAD BENDS PART BEFORE DIE MOUNTED CAM ENGAGES

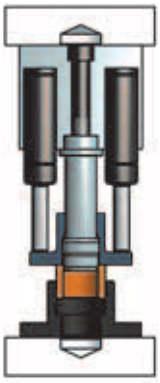


SPRING PAD HOLDS PART WHILE CAM ACTION FLANGES PART

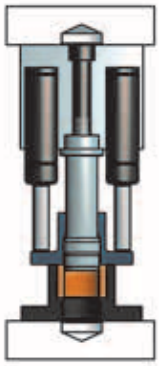


Bushing Installation and Sizing

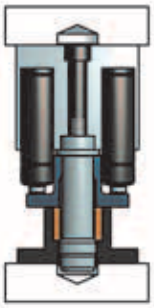
PART LOADED AND MICRO SPRING PRESSING BUSHING



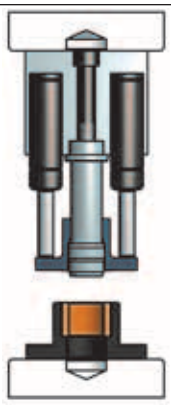
BUSHING PRESSED INTO PLACE



BUSHING SIZED AND INSTALLED



PART COMPLETE UNLOAD



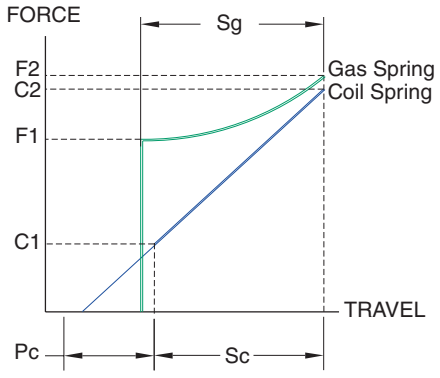
The above examples are conceptual and are not intended as engineering design for specific forces or applications. All tool designs must be individually engineered for their intended function. See pages 19-20 for recommendations and limitations for mounting and installation.

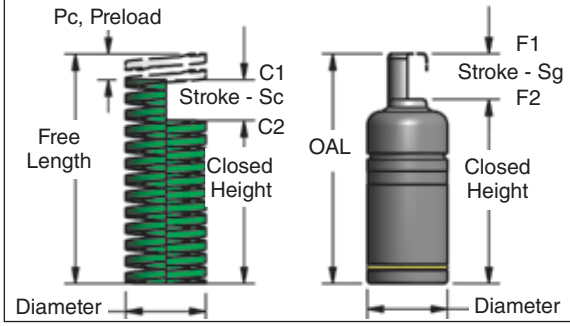
Coil Spring Conversion

Unlike coil springs, DADCO Micro Nitrogen Gas Springs provide full rated force on contact. This force is repeatable, eliminating scrap and maximizing productivity. One Micro Spring can provide the force of several heavy duty coil springs, and will outlast the coil springs.

Most die springs have a limitation on compression, after which there is failure or severely reduced cycle life. Heavy and extra heavy duty die springs can only be compressed 15-20% of the closed height, after which there is failure or severely reduced cycle life.

Coil spring force is based on the spring rate of the coil spring. Spring rate is determined by the material, wire diameter, spring diameter, number of coils and height of the coil spring.

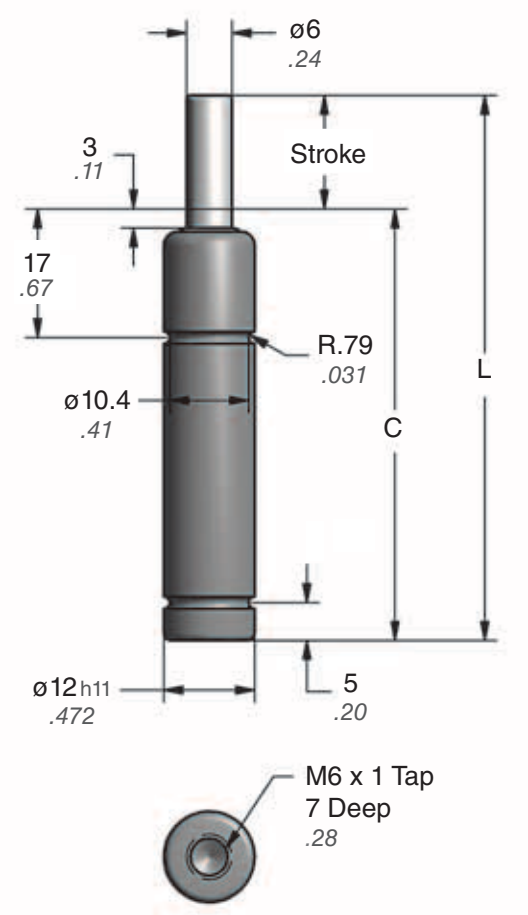




All coil springs require a preload to achieve a contact force (C1) larger than zero. For most spring applications the preload force is the force required to strip, hold, form or return the part. As a result of the spring rate, coil springs have a continuing increase in force after preload (C1 - C2).

DADCO Micro Series Gas Springs have a much flatter curve that will not exceed 30% of the original force for any stroke size (F1 - F2). When retrofitting coil springs calculate the total preloaded force required for the application. The number and color (load rating) of Micro Gas Springs can then be determined.

Micro 45®



Part No.	Stroke mm *inch*	C	L ±0.4 *±0.015*
•C.045.007	07 *.28*	49 *1.93*	56 *2.205*
C.045.010	10 *.39*	52 *2.05*	62 *2.441*
C.045.013	12.7 *.50*	54.7 *2.15*	67.4 *2.654*
•C.045.015	15 *.59*	57 *2.24*	72 *2.835*
•C.045.025	25 *.98*	67 *2.64*	92 *3.622*
•C.045.038	38 *1.50*	80 *3.15*	118 *4.646*
•C.045.050	50 *1.97*	92 *3.62*	142 *5.591*
C.045.063	63.5 *2.50*	108.5 *4.27*	172 *6.772*
C.045.080	80 *3.15*	125 *4.92*	205 *8.071*

• *Preferred Sizes*

Force Chart	Initial lb. *daN*	Final lb. *daN*	Pressure psi *bar*
Yellow - YW	112 *50*	166 *74*	2560 *177*
Red - RD	84 *37*	124 *55*	1920 *132*
Blue - BU	56 *25*	83 *37*	1280 *88*
Green - GR	28 *12*	41 *18*	640 *44*
Black - BK	See Charts		

On-Contact Force – Adjustable Black Model

Imperial

Pressure (psi)	Force (lb.-f)
2560	112
2200	96
2000	88
1750	77
1500	66
1000	44
500	22
260	11

P = F ÷ .044 F = P x .044

Metric

Pressure (bar)	Force (daN)
177	50
150	42
125	35
100	28
75	21
50	14
35	10
18	5

P = F ÷ .283 F = P x .283

Ordering Example: C.045.007. *GR*

Part Number:
Includes Series, Model and Stroke Length

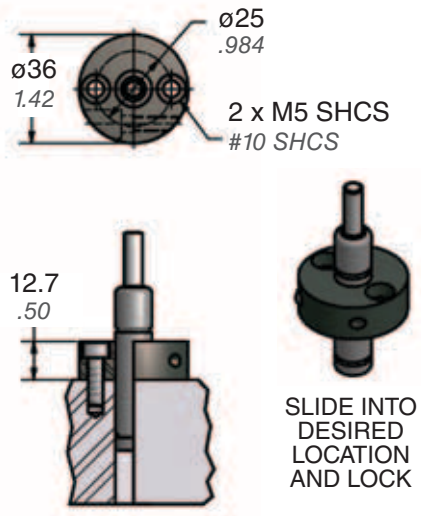
Force:
YW, RD, BU, GR
BK – Black adjustable model - specify pressure:
18 – 177 bar *(260 – 2560 psi)*.
Ordering Example: C.045.007.BK.150

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Micro 45® Mounts

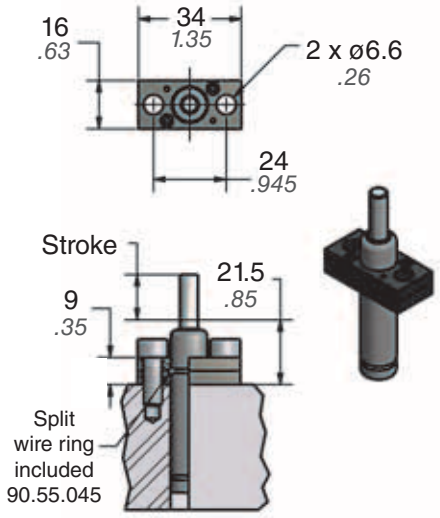
Attachable Mount Options

RM — C45-RM



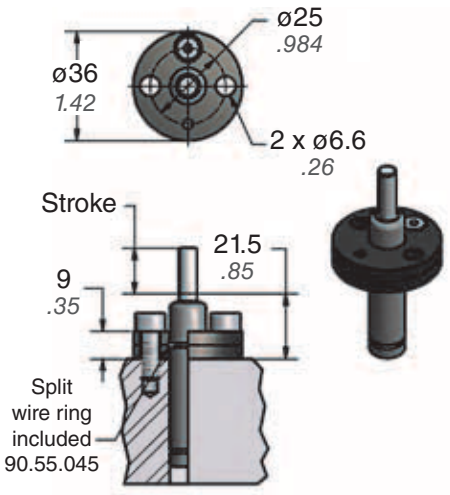
DADCO-LOK

NF — C45-NF



Narrow Flange

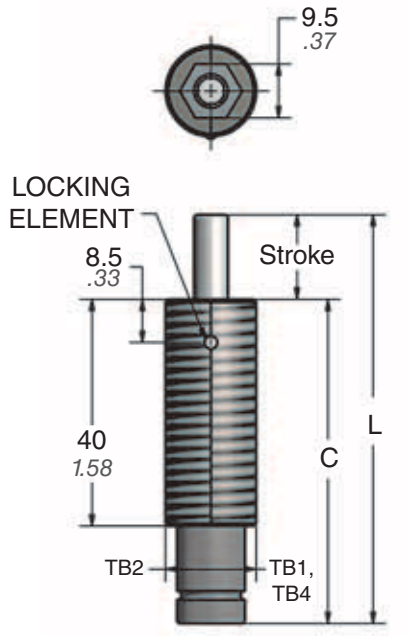
RF — C45-RF



Round Flange

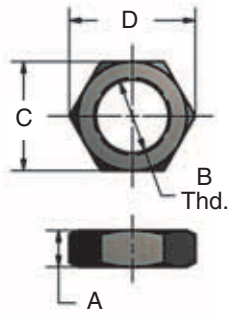
Threaded Body Styles

TB1, TB2 & TB4



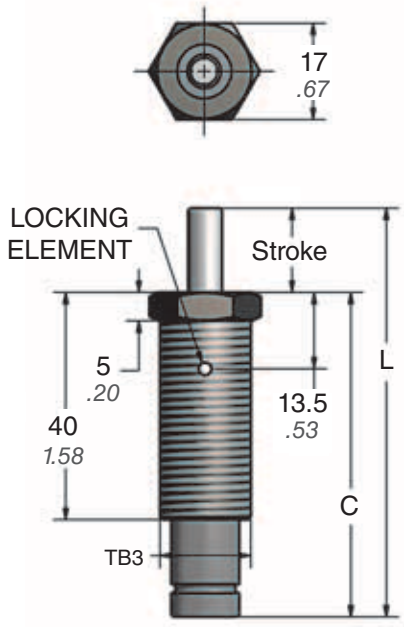
	TB2	TB4	TB1
Thread	M16 x 1.5	M16 x 2	5/8˝-11

Jam Nut



	C45-JN1	SL16-JN	C45-JN4
A	9 .36	8 .31	8 .31
B	5/8˝-11	M16 x 1.5	M16 x 2
C	27 1.06	24 .95	24 .95
D	31 1.23	28 1.10	28 1.10

TB3



	TB3
Thread	M16 x 2

Ordering Example:

C.045.007. *TB1.* *GR*

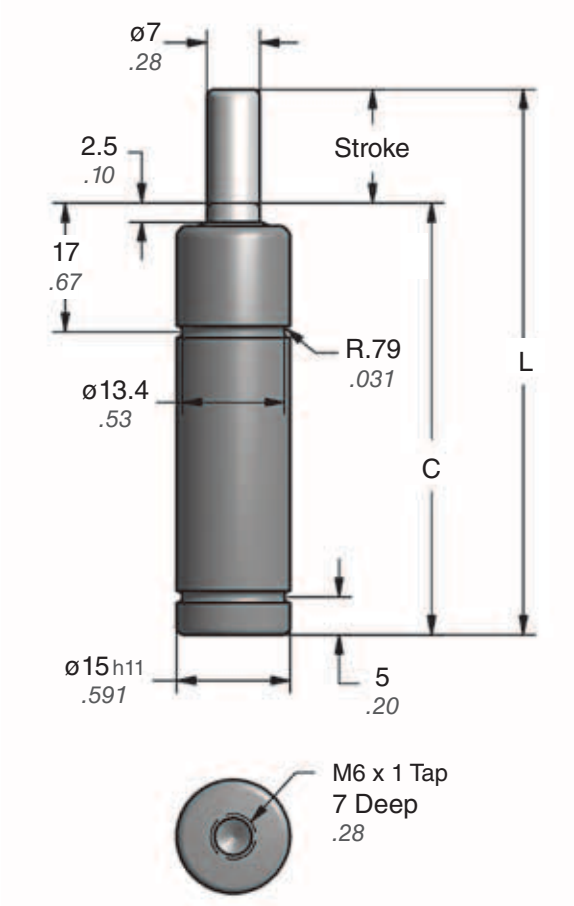
Part Number:
Includes Series, Model and Stroke Length

Mount Option:
RM, NF, RF, TB1, TB2, TB3, TB4
Mount Only Ordering Example: C45-RM

Force:
YW, RD, BU, GR
BK – Black adjustable model - specify pressure:
18 – 177 bar *(260 – 2560 psi)*.
Ordering Example: C.045.007.TB1.BK.150

Refer to Bulletin #B04120 for GC.045.015.TB5 information.

Micro 70®



Part No.	Stroke mm *inch*	C	L ±0.4 *±0.015*
•C.070.007	07 *.28*	49 *1.93*	56 *2.205*
C.070.010	10 *.39*	52 *2.05*	62 *2.441*
C.070.013	12.7 *.50*	54.7 *2.15*	67.4 *2.654*
•C.070.015	15 *.59*	57 *2.24*	72 *2.835*
•C.070.025	25 *.98*	67 *2.64*	92 *3.622*
•C.070.038	38 *1.50*	80 *3.15*	118 *4.646*
•C.070.050	50 *1.97*	92 *3.62*	142 *5.591*
C.070.063	63.5 *2.50*	108.5 *4.27*	172 *6.772*
•C.070.080	80 *3.15*	125 *4.92*	205 *8.071*
C.070.100	100 *3.94*	145 *5.71*	245 *9.646*
C.070.125	125 *4.92*	170 *6.69*	295 *11.614*

•*Preferred Sizes*

Force Chart	Initial lb. *daN*	Final lb. *daN*	Pressure psi *bar*
Yellow - YW	154 *68*	208 *93*	2560 *177*
Red - RD	115 *51*	156 *69*	1920 *132*
Blue - BU	77 *34*	104 *46*	1280 *88*
Green - GR	38 *17*	52 *23*	640 *44*
Black - BK	See Charts		

On-Contact Force – Adjustable Black Model

Imperial

Pressure (psi)	Force (lb.-f)
2560	154
2200	132
2000	120
1750	105
1500	90
1000	60
500	30

P = F ÷ .060 F = P x .060

Metric

Pressure (bar)	Force (daN)
177	68
150	57
125	48
100	38
75	29
50	19
35	13

P = F ÷ 0.38 F = P x 0.38

Ordering Example: C.070.007. *GR*

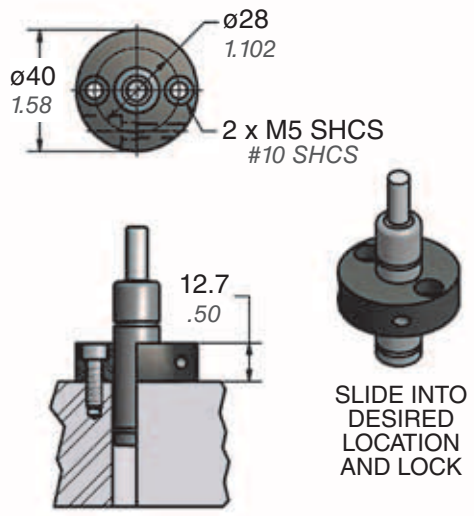
Part Number:
Includes Series, Model and Stroke Length

Force:
YW, RD, BU, GR
BK – Black adjustable model - specify pressure:
35 – 177 bar *(500 – 2560 psi)*.
Ordering Example: C.070.007.BK.150

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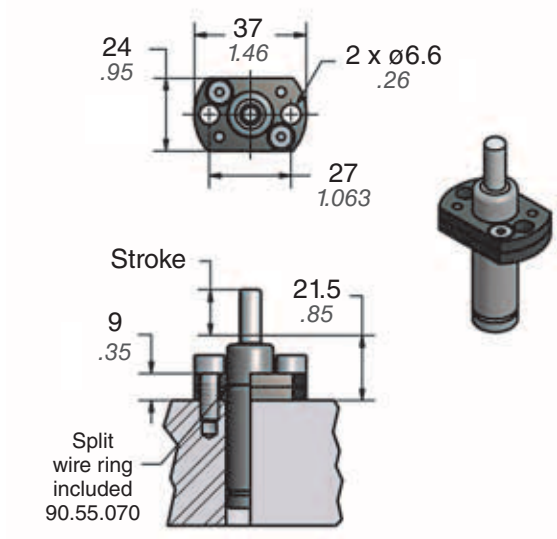
Micro 70® Mounts

RM **C70-RM**



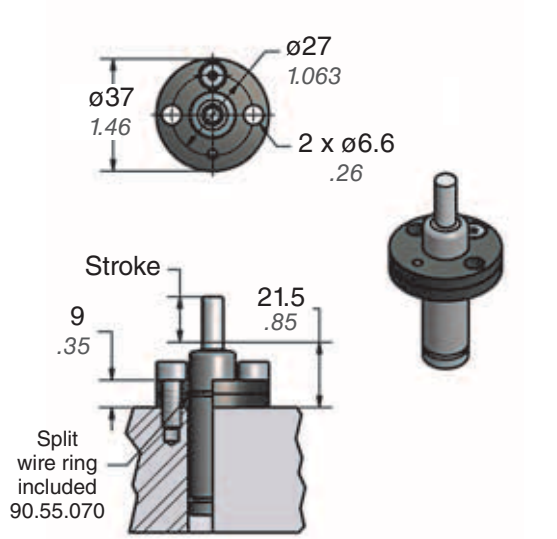
DADCO-LOK

FA **C70-FA**



Narrow Flange

RF **C70-RF**



Round Flange

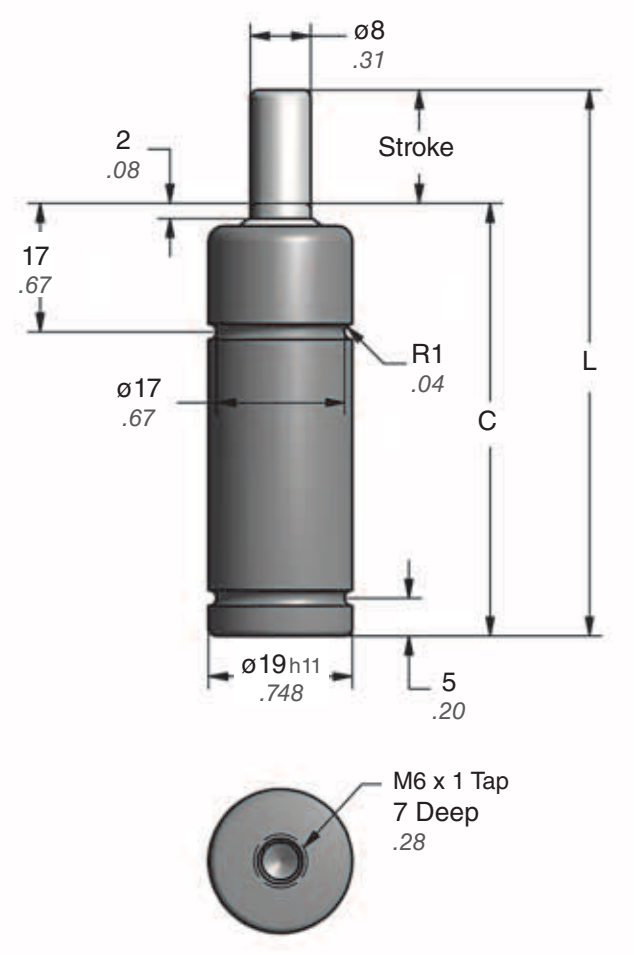
Ordering Example: C.070.007. *RM.* *GR*

Part Number:
Includes Series, Model and Stroke Length

Mount Option:
RM, FA, RF
Mount Only Ordering Example: *C70-RM*

Force:
YW, RD, BU, GR
BK – Black adjustable model - specify pressure:
35 – 177 bar *(500 – 2560 psi)*.
Ordering Example: *C.070.007.RM.BK.150*

Micro 90®



Part No.	Stroke mm *inch*	C	L ±0.4 *±0.015*
•C.090.007	07 *.28*	49 *1.93*	56 *2.205*
C.090.010	10 *.39*	52 *2.05*	62 *2.441*
C.090.013	12.7 *.50*	54.7 *2.15*	67.4 *2.654*
•C.090.015	15 *.59*	57 *2.24*	72 *2.835*
•C.090.025	25 *.98*	67 *2.64*	92 *3.622*
•C.090.038	38 *1.50*	80 *3.15*	118 *4.646*
•C.090.050	50 *1.97*	92 *3.62*	142 *5.591*
C.090.063	63.5 *2.50*	108.5 *4.27*	172 *6.772*
•C.090.080	80 *3.15*	125 *4.92*	205 *8.071*
C.090.100	100 *3.94*	145 *5.71*	245 *9.646*
C.090.125	125 *4.92*	170 *6.69*	295 *11.614*
C.090.150	150 *5.91*	203 *7.99*	353 *13.898*
C.090.160	160 *6.30*	213 *8.39*	373 *14.685*
C.090.175	175 *6.89*	228 *8.98*	403 *15.866*
C.090.200	200 *7.87*	253 *9.96*	453 *17.835*

•*Preferred Sizes*

Force Chart	Initial lb. *daN*	Final lb. *daN*	Pressure psi *bar*
Yellow - YW	200 *89*	256 *114*	2560 *177*
Red - RD	150 *66*	192 *85*	1920 *132*
Blue - BU	100 *44*	128 *57*	1280 *88*
Green - GR	50 *22*	64 *28*	640 *44*
Purple - PR	20 *9*	26 *12*	260 *18*
Orange - OR	10 *5*	13 *6*	130 *9*
Black - BK	See Charts		

On-Contact Force – Adjustable Black Model

Imperial

Pressure (psi)	Force (lb.-f)
2560	200
2200	172
2000	156
1750	136
1500	117
1000	78
500	39

P = F ÷ .078 F = P x .078

Metric

Pressure (bar)	Force (daN)
177	89
150	75
125	63
100	50
75	38
50	25
35	17

P = F ÷ 0.50 F = P x 0.50

Ordering Example: C.090.007. GR

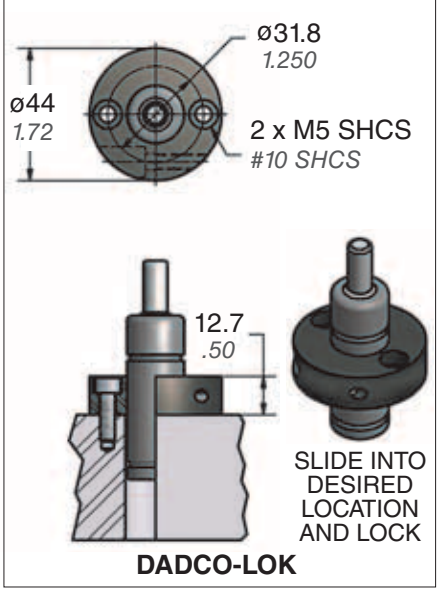
Part Number:
Includes Series, Model and Stroke Length
150 mm - 200 mm strokes; contact DADCO for application evaluation.

Force:
YW, RD, BU, GR, PR, OR
BK – Black adjustable model - specify pressure: 35 – 177 bar *(500 – 2560 psi)*.
Ordering Example: C.090.007.BK.150

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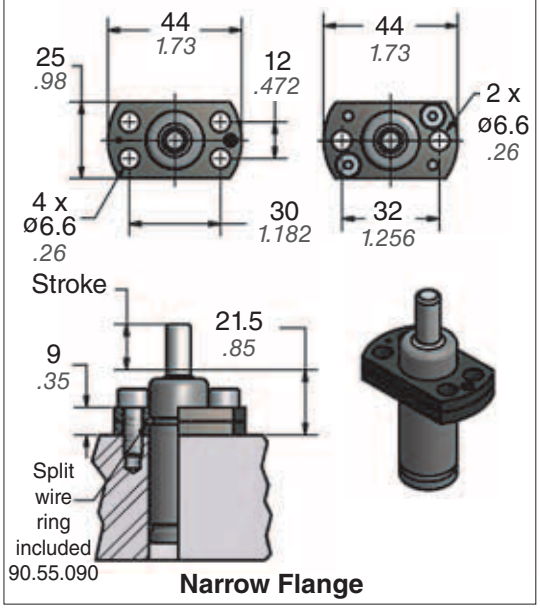
Attachable Mount Options

RM — C90-RM



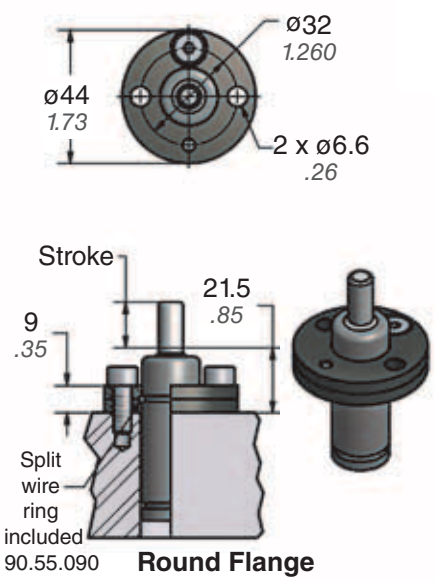
DADCO-LOK

FA / C90-FA — VFA / C90-VFA



Narrow Flange

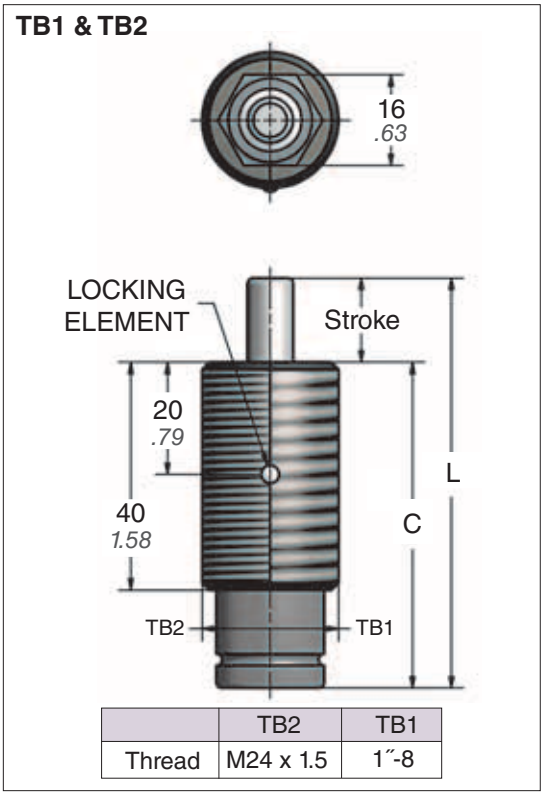
RF — C90-RF



Round Flange

Threaded Body Styles

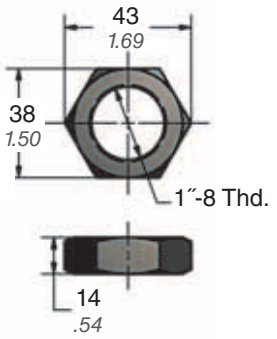
TB1 & TB2



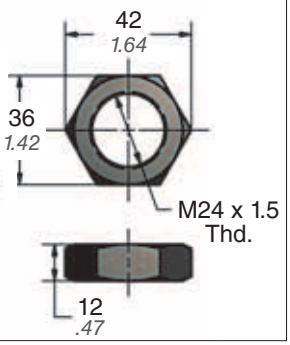
	TB2	TB1
Thread	M24 x 1.5	1″-8

Jam Nut

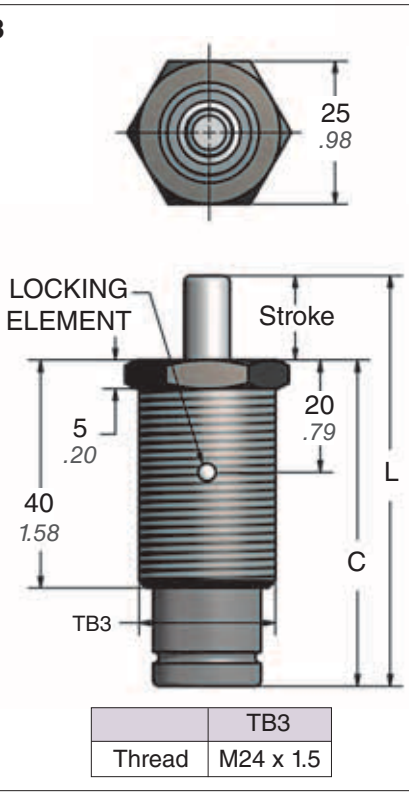
C90-JN1 (1″-8 Thd.)



C90-JN2 (M24 x 1.5 Thd.)



TB3



	TB3
Thread	M24 x 1.5

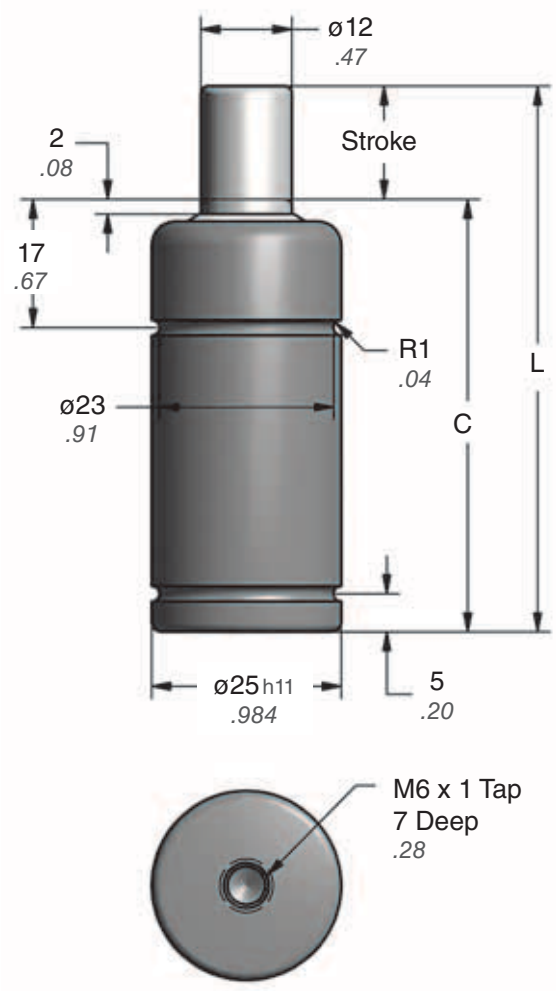
Ordering Example: C.090.007. *TB1.* *GR*

Part Number:
Includes Series, Model and Stroke Length

Mount Option:
RM, FA, VFA, RF, TB1, TB2, TB3
Mount Only Ordering Example: C90-RM

Force:
YW, RD, BU, GR, PR, OR
BK – Black adjustable model - specify pressure:
35 – 177 bar *(500 – 2560 psi)*.
Ordering Example: C.090.007.TB1.BK.150

Micro 180®



Part No.	Stroke mm *inch*	C	L ±0.4 *±0.015*
•C.180.007	07 *.28*	49 *1.93*	56 *2.205*
C.180.010	10 *.39*	52 *2.05*	62 *2.441*
C.180.013	12.7 *.50*	54.7 *2.15*	67.4 *2.654*
•C.180.015	15 *.59*	57 *2.24*	72 *2.835*
•C.180.025	25 *.98*	67 *2.64*	92 *3.622*
•C.180.038	38 *1.50*	80 *3.15*	118 *4.646*
•C.180.050	50 *1.97*	92 *3.62*	142 *5.591*
C.180.063	63.5 *2.50*	108.5 *4.27*	172 *6.772*
•C.180.080	80 *3.15*	125 *4.92*	205 *8.071*
C.180.100	100 *3.94*	145 *5.71*	245 *9.646*
C.180.125	125 *4.92*	170 *6.69*	295 *11.614*
C.180.150	150 *5.91*	203 *7.99*	353 *13.898*
C.180.160	160 *6.30*	213 *8.39*	373 *14.685*
C.180.175	175 *6.89*	228 *8.98*	403 *15.866*
C.180.200	200 *7.87*	253 *9.96*	453 *17.835*

•*Preferred Sizes*

Force Chart	Initial lb. *daN*	Final lb. *daN*	Pressure psi *bar*
Yellow - YW	450 *200*	612 *272*	2560 *177*
Red - RD	337 *149*	459 *204*	1920 *132*
Blue - BU	224 *100*	306 *136*	1280 *88*
Green - GR	112 *50*	153 *68*	640 *44*
Black - BK	See Charts		

On-Contact Force – Adjustable Black Model

Imperial

Pressure (psi)	Force (lb.-f)
2560	450
2200	387
2000	351
1750	307
1500	263
1000	175
500	88

P = F ÷ .175 F = P x .175

Metric

Pressure (bar)	Force (daN)
177	200
150	170
125	141
100	113
75	85
50	57
35	39

P = F ÷ 1.13 F = P x 1.13

Ordering Example: C.180.007. *GR*

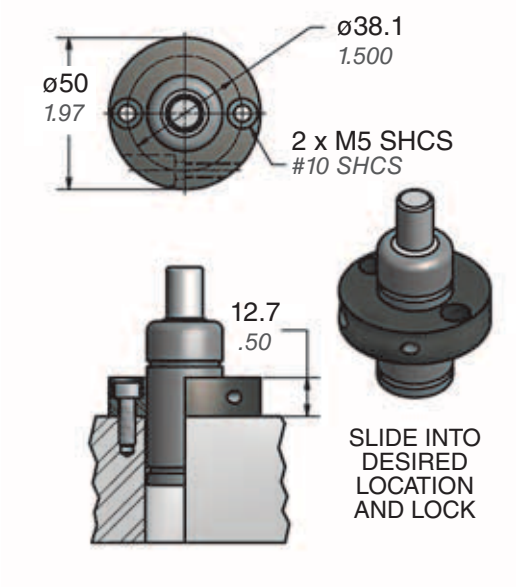
Part Number:
Includes Series, Model and Stroke Length
150 mm - 200 mm strokes; contact DADCO for application evaluation.

Force:
YW, RD, BU, GR
BK – Black adjustable model - specify pressure: 35 – 177 bar *(500 – 2560 psi)*.
Ordering Example: C.180.007.BK.150

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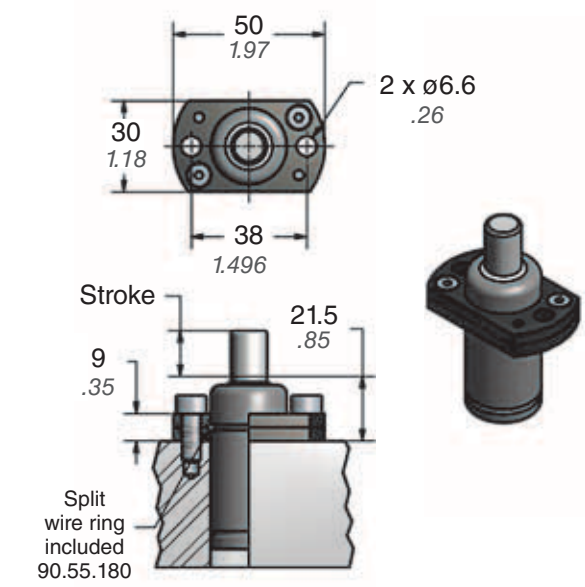
Micro 180® Mounts

RM **C180-RM**



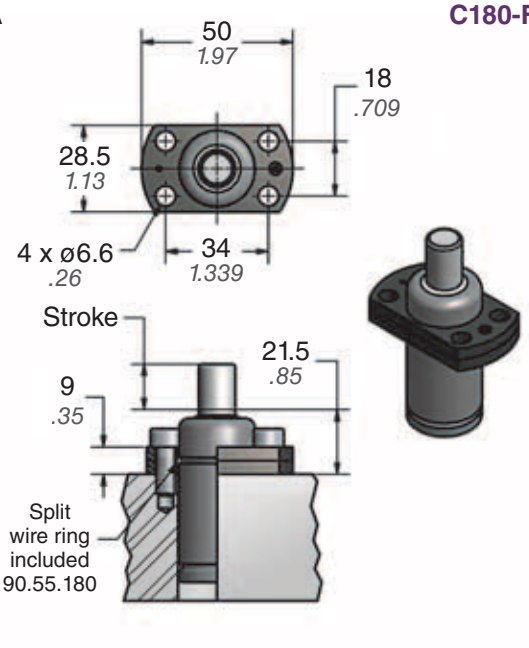
DADCO-LOK

VFA **C180-VFA**



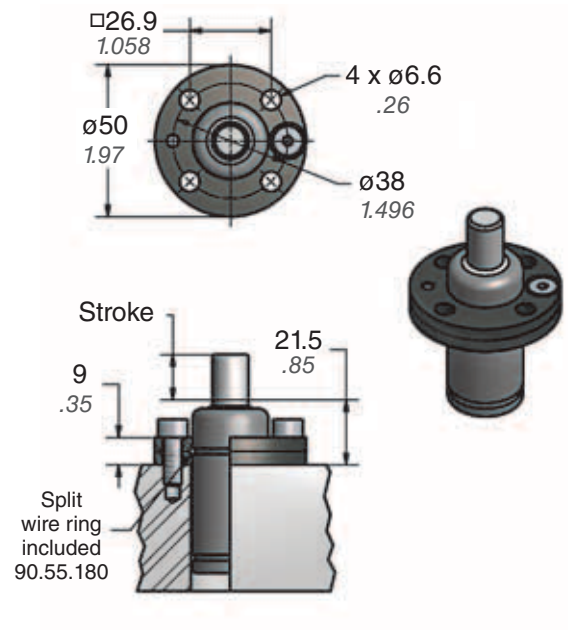
Narrow Flange

FA **C180-FA**



Narrow Flange

RF **C180-RF**



Round Flange

Ordering Example:

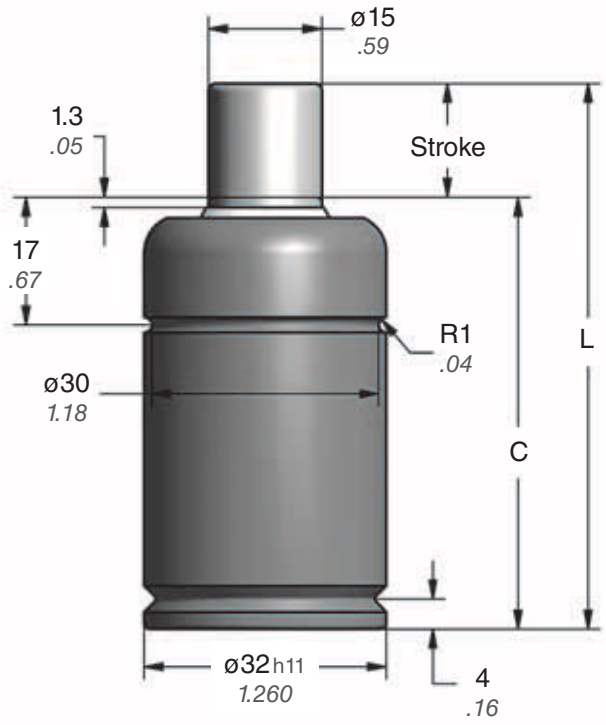
C.180.007. *RM.* *GR*

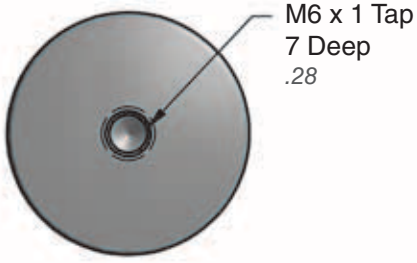
Part Number:
Includes Series, Model and Stroke Length

Mount Option:
RM, FA, VFA, RF
Mount Only Ordering Example: C180-RM

Force:
YW, RD, BU, GR
BK – Black adjustable model - specify pressure:
35 – 177 bar *(500 – 2560 psi)*.
Ordering Example: C.180.007.RM.BK.150

Micro 250®





Part No.	Stroke mm *inch*	C	L ±0.4 *±0.015*
•C.250.007	07 *.28*	49 *1.93*	56 *2.205*
C.250.010	10 *.39*	52 *2.05*	62 *2.441*
C.250.013	12.7 *.50*	54.7 *2.15*	67.4 *2.654*
•C.250.015	15 *.59*	57 *2.24*	72 *2.835*
•C.250.025	25 *.98*	67 *2.64*	92 *3.622*
•C.250.038	38 *1.50*	80 *3.15*	118 *4.646*
•C.250.050	50 *1.97*	92 *3.62*	142 *5.591*
C.250.063	63.5 *2.50*	108.5 *4.27*	172 *6.772*
•C.250.080	80 *3.15*	125 *4.92*	205 *8.071*
C.250.100	100 *3.94*	145 *5.71*	245 *9.646*
C.250.125	125 *4.92*	170 *6.69*	295 *11.614*

• *Preferred Sizes*

Force Chart	Initial lb. *daN*	Final lb. *daN*	Pressure psi *bar*
Yellow - YW	701 *313*	940 *418*	2560 *177*
Red - RD	526 *233*	705 *314*	1920 *132*
Blue - BU	351 *156*	470 *209*	1280 *88*
Green - GR	175 *78*	235 *105*	640 *44*
Black - BK	See Charts		

On-Contact Force – Adjustable Black Model

Imperial

Pressure (psi)	Force (lb.-f)
2560	701
2200	603
2000	548
1750	479
1500	411
1000	274
500	137

P = F ÷ .274 F = P x .274

Metric

Pressure (bar)	Force (daN)
177	313
150	265
125	221
100	177
75	133
50	88
35	60

P = F ÷ 1.77 F = P x 1.77

Ordering Example: C.250.007. *GR*

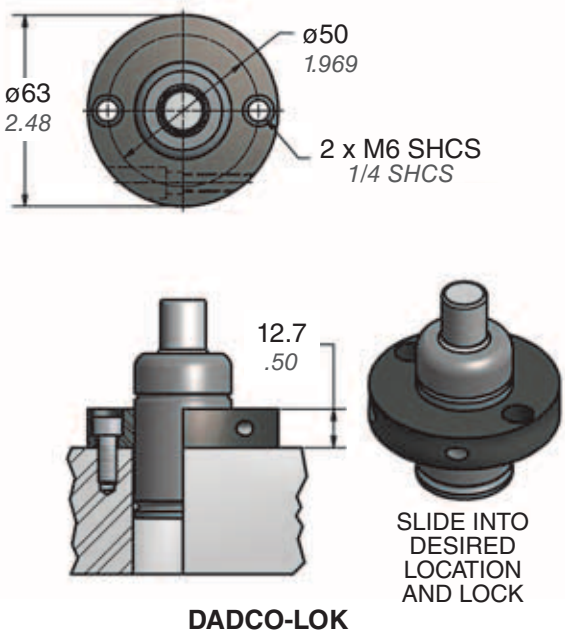
Part Number:
Includes Series, Model and Stroke Length

Force:
YW, RD, BU, GR
BK – Black adjustable model - specify pressure:
35 – 177 bar *(500 – 2560 psi)*.
Ordering Example: C.250.007.BK.150

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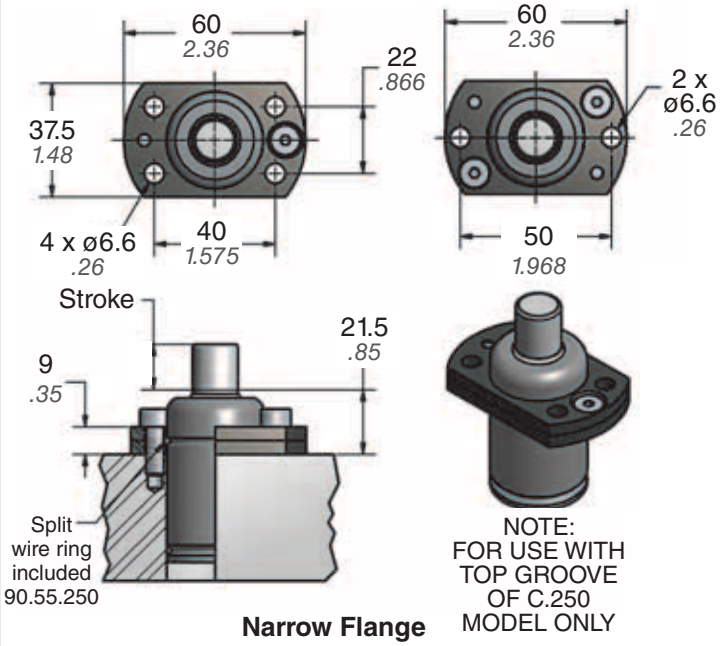
Micro 250® Mounts

RM — C250-RM



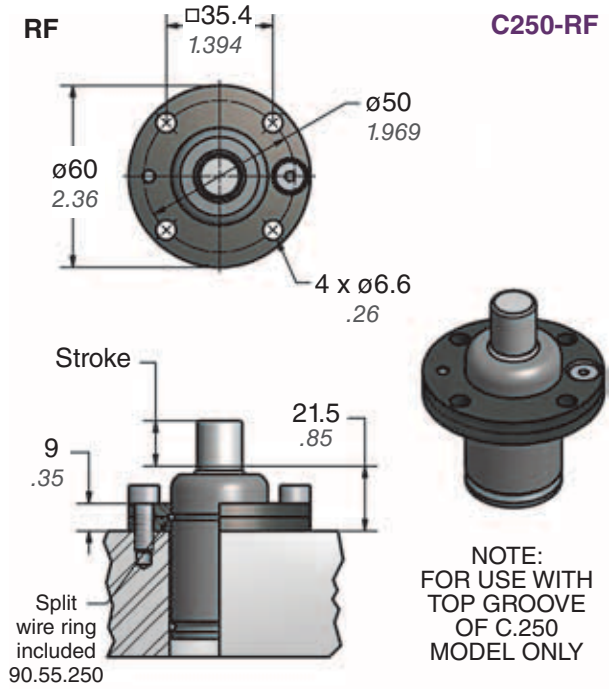
DADCO-LOK

FA / C250-FA — VFA / C250-VFA



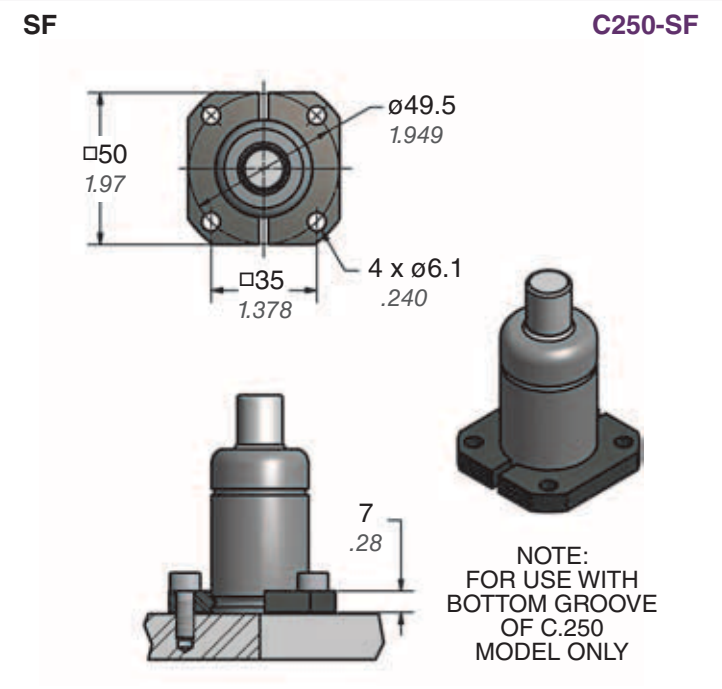
Narrow Flange

RF — C250-RF



Round Flange

SF — C250-SF



Split Flange

Ordering Example:

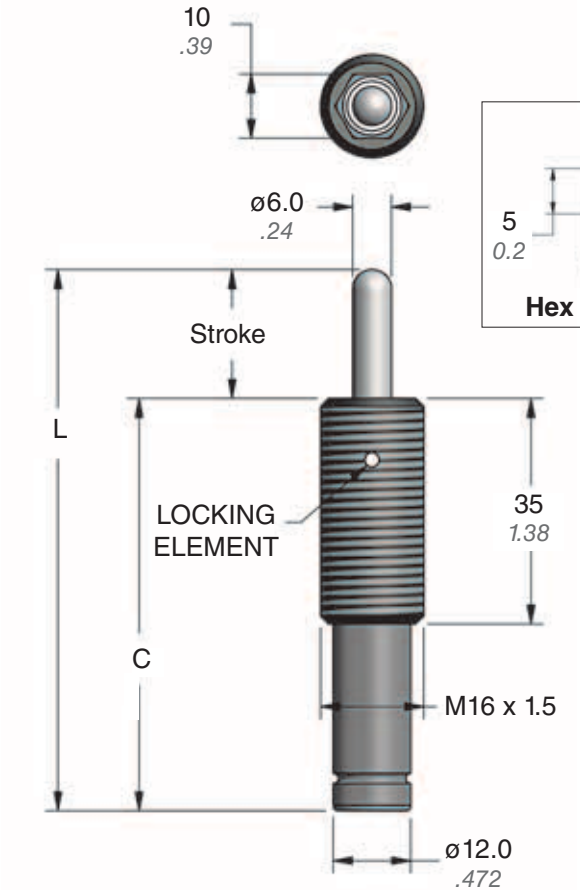
C.250.007. *RM.* *GR*

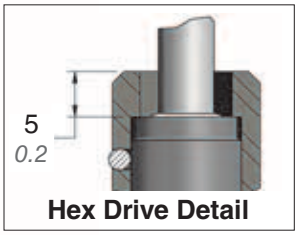
Part Number:
Includes Series, Model and Stroke Length

Mount Option:
RM, FA, VFA, RF, SF
Mount Only Ordering Example: C250-RM

Force:
YW, RD, BU, GR
BK – Black adjustable model - specify pressure:
35 – 177 bar *(500 – 2560 psi)*.
Ordering Example: C.250.007.RM.BK.150

E.16 Stock Lifter

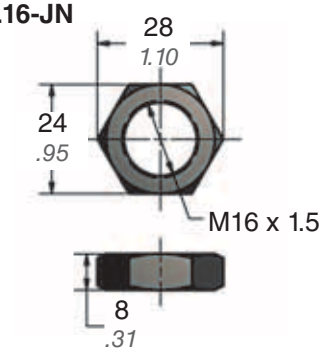




Part No.	Stroke mm *inch*	C	L ±0.4 *±0.015*
E.16.020	20 *.79*	64 *2.52*	84 *3.31*
E.16.050	50 *1.97*	94 *3.70*	144 *5.67*
E.16.080	80 *3.15*	127 *5.00*	207 *8.15*

DADCO's E.16 Nitrogen Gas Stock Lifter matches the European VDI-BAK standard and the Ford WDX35-70 standard.

Jam Nut
SL16-JN



On-Contact Force

Imperial

Pressure (psi)	Force (lb.-f)
2175	95
1088	48
580	25
290	13

P = F ÷ .044 F = P x .044

Metric

Pressure (bar)	Force (daN)
150	42
75	21
40	11
20	6

P = F ÷ .283 F = P x .283

Ordering Example:

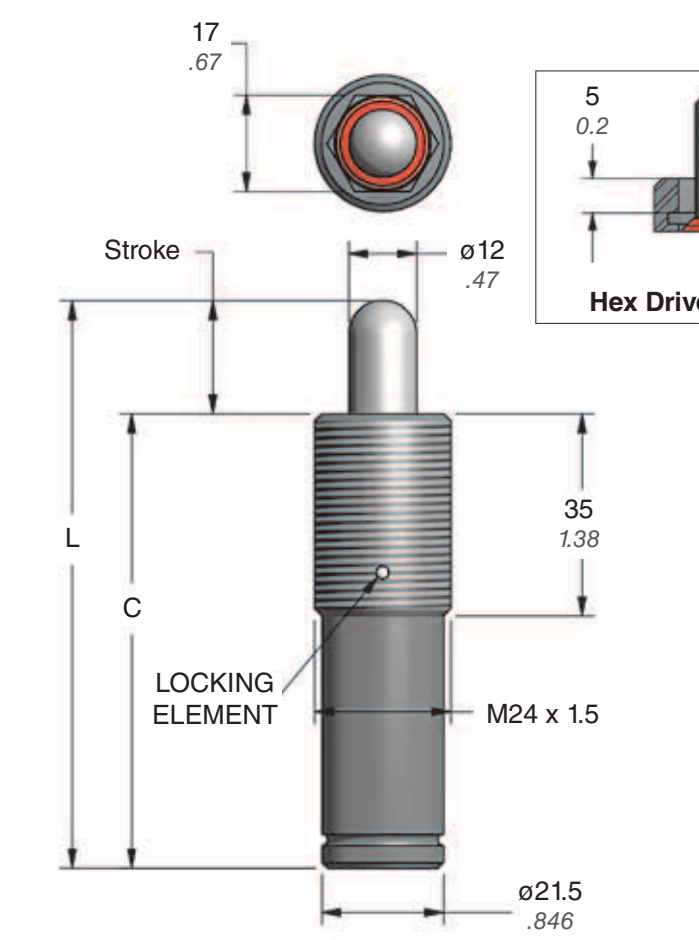
E.16.020. *150*

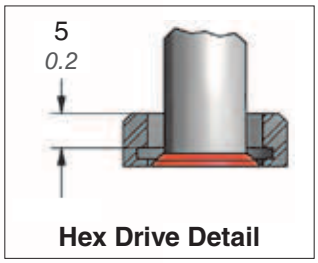
Part Number:
Includes Series, Model and Stroke Length

Charging Pressure:
Specify pressure: 20 – 150 bar *(290 psi – 2175 psi)*.
When not specified, default is 150 bar.

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E.24 Stock Lifter

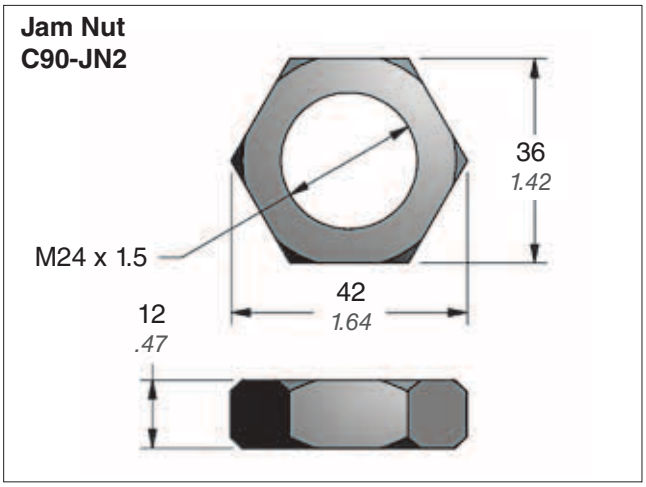




Part No.	Stroke mm *inch*	C	L ±0.4 *±0.015*
E.24.020	20 *.79*	80 *3.15*	100 *3.94*
E.24.050	50 *1.97*	110 *4.33*	144 *6.30*
E.24.080	80 *3.15*	140 *5.51*	220 *8.66*

DADCO's E.24 Nitrogen Gas Stock Lifter matches the European VDI-BAK standard and the Ford WDX35-70 standard.

Jam Nut
C90-JN2



On-Contact Force

Imperial

Pressure (psi)	Force (lb.-f)
2175	381
1088	191
580	102
290	51

P = F ÷ .175 F = P x .175

Metric

Pressure (bar)	Force (daN)
150	170
75	85
40	45
20	23

P = F ÷ 1.13 F = P x 1.13

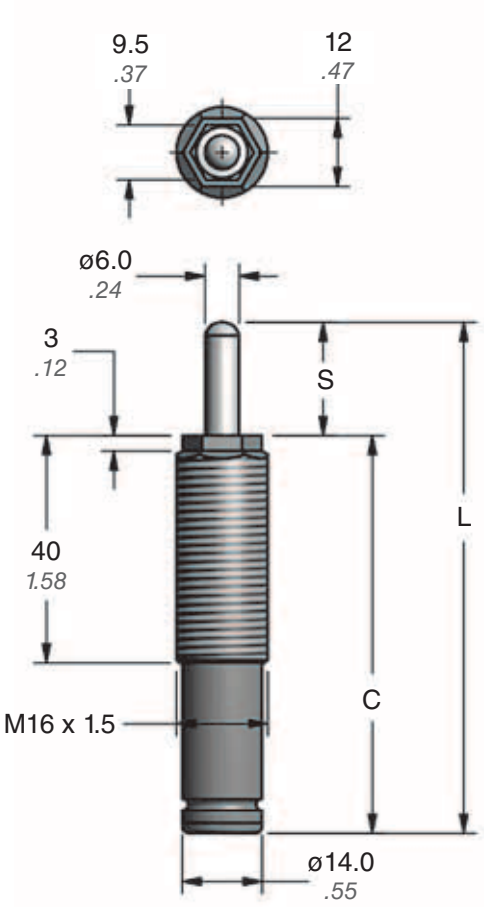
Ordering Example:

E.24.020. *150*

Part Number:
Includes Series, Model and Stroke Length

Charging Pressure:
Specify pressure: 20 – 150 bar *(290 psi – 2175 psi)*.
When not specified, default is 150 bar.

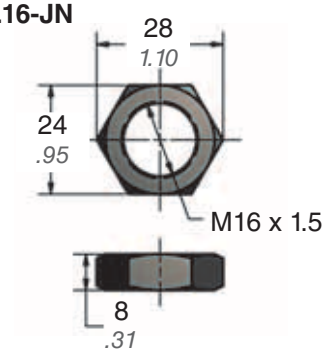
SL.16 Stock Lifter



Part No.	Stroke mm *inch*	C	L ±0.4 *±0.015*
SL.16.010	10 *.39*	70 *2.76*	80 *3.15*
• SL.16.020	20 *.79*	80 *3.15*	100 *3.94*
SL.16.030	30 *1.18*	90 *3.54*	120 *4.72*
SL.16.040	40 *1.57*	100 *3.94*	140 *5.51*
• SL.16.050	50 *1.97*	110 *4.33*	160 *6.30*
SL.16.060	60 *2.36*	120 *4.72*	180 *7.09*
SL.16.070	70 *2.76*	130 *5.12*	200 *7.87*
• SL.16.080	80 *3.15*	140 *5.51*	220 *8.66*
SL.16.100	100 *3.94*	160 *6.30*	260 *10.24*

• *Preferred Sizes*

Jam Nut
SL16-JN



On-Contact Force

Imperial

Pressure (psi)	Force (lb.-f)
2611	114
2176	95
1088	48
580	25
290	13

P = F ÷ .044 F = P x .044

Metric

Pressure (bar)	Force (daN)
180	51
150	42
75	21
40	11
20	6

P = F ÷ .283 F = P x .283

Ordering Example:

SL.16.020. *150*

Part Number:
Includes Series, Model and Stroke Length

Charging Pressure:
Specify pressure: 20 – 180 bar *(290 psi – 2611 psi)*.
When not specified, default is 150 bar.

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Tools & Accessories

Micro Load Cell
LC-____ (045, 090, 180 or 250)

Use the Micro Load Cell with a Micro Test Stand or arbor press to determine the force of a Micro Spring. Depress the micro rod 1/16˝ to read gas spring force from the color-coded gauge. Request Bulletin No. B07108 for additional information.



Micro Test Stand
MTS-125

Use the Micro Test Stand and Load Cell for precise measurement of gas spring force on contact. Request Bulletin No. B01127 for additional information.



RT-24-A (for use with E.24)
RT-90-A (for use with **Micro 90**® TB1 and TB2**)**

When placed over the piston rod, the installation and removal tool engages the hex socket for easy installation and removal of threaded body micros.



RT-Ratcheting Tool
RT-45 (for use with **Micro 45**® TB1, TB2 and TB4)
RT-16-B (for use with **Micro 90**® TB1 and TB2)
RT-16-D (for use with SL.16)
RT-16-F (for use with E.16)

Ratcheting tool with internal hex drive for easy installation and removal of threaded body Micros. For complete list of Ratcheting Tools refer to Bulletin No. B04139A.



Specialized Mounts

For customers with special applications that have space constraints or require return force, DADCO offers custom mount options. Contact DADCO for additonal information.



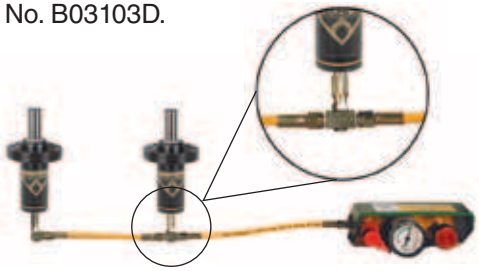
Micro Wiper Cap

For customers with applications where aggressive draw die compound is used, DADCO offers the Micro Wiper Cap. The wiper cap can be ordered in a variety of materials and is installed at the factory to guard against draw die contamination, request Bulletin No. B03102A. Alternatively, DADCO offers an internal wiper in different materials. Contact DADCO for additional information.



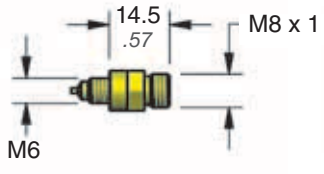
Linked Micro System

Typically DADCO Micro Series Nitrogen Gas Springs are operated self-contained, however they may also be linked. When operated as a linked system, adjustment, monitoring, draining and refilling can be performed from a central control panel mounted outside the die, request Bulletin No. B03103D.

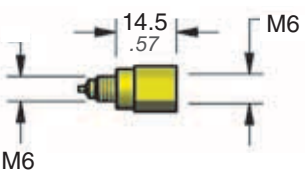


Micro Series Port Adapters

DADCO's Micro Series Port Adapters are designed specifically to work with DADCO's Micro Series Nitrogen Gas Springs manufactured after August 1, 2003. These port adapters may be used with DADCO's *MINIFLEX*® Hose and Fittings, refer to Catalog No. C09118C.



90.607.122
(L-122)
Micro Service Fitting



90.607.038
(L-38)
Micro Port Adapter Extension

Charging Information

CAUTION:
Always wear safety goggles when performing any maintenance work on gas springs.

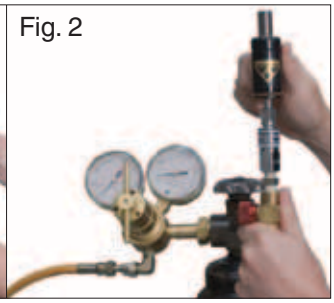
Charging Micro Gas Springs

- When filling the Micro Spring, initially fill with low pressure (< 4 bar or 70 psi) to extend rod fully; then fill to desired pressure. Hold the spring vertically at all times during filling *(Fig. 1)*.
- The Micro Spring charging pressure range varies by gas spring model. Verify range before charging.
- ***All Micro Springs should be inspected before recharging.***
- ***Do not recharge gas springs if damaged. Refer to discharging instructions below for proper disposal.***
- Use the Quick Disconnect Charging Nipple and the High Pressure Quick Disconnect Charging Assembly to charge the Micro Spring to the appropriate pressure *(Fig. 2)*.

Fig. 1



Fig. 2



Quick Disconnect Charging Nipple
90.310.143 *(CN-4)*

Use the DADCO Quick Disconnect Charging Nipple to charge the Micro Series Gas Spring. For more information contact DADCO.



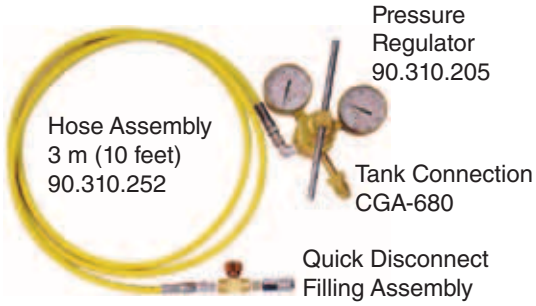
DADCO Charging Adapter
90.315.5

Use the DADCO Charging Adapter to easily charge and discharge pressure in DADCO's Micro Series Gas Spring. *Not recommended for checking pressure due to small size of Micro Series Gas Springs.*



High Pressure Quick Disconnect Charging Assembly
90.310.041 *(CA-41)*

Use the DADCO High Pressure Quick Disconnect Charging Assembly, 90.310.041, with the 90.310.143 Charging Nipple or 90.315.5 Charging Adapter to charge self-contained gas springs. The 90.310.041 includes the 90.310.205 Pressure Regulator, 90.310.252 Hose Assembly and the 90.310.338 Quick Disconnect Filling Assembly. *A standard pressure charging assembly, 90.310.040 is available for pressures below 2200 psi, for a CGA-580 tank. For more information contact DADCO.*



How to discharge a Micro Gas Spring before disposal

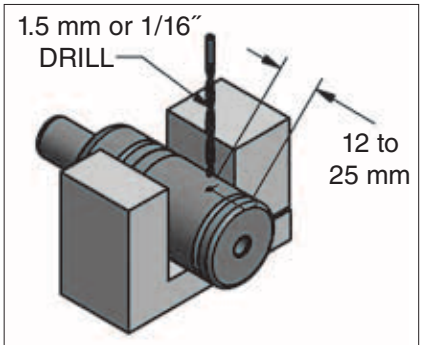
CAUTION: Before disposing of damaged or worn out gas springs be sure to discharge all pressure. Contact DADCO for additional information.

1. Discharge through the adjustable valve using the Valve Bleed Tool or Charging Adapter, 90.315.5.



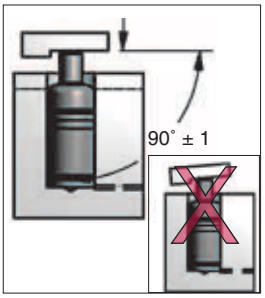
Valve Bleed Tool
90.360.4

2. If spring is damaged and cannot be discharged using the Valve Bleed Tool then drill a hole to discharge.



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General Recommendations



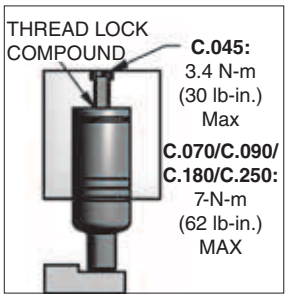
Side loading from axial or contact misalignment should be minimized, <1˚.



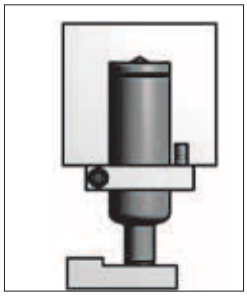
It is necessary to have a flat surface against the base of the spring in all circumstances. Incorrect pockets may cause structural damage or reduced life.



All properly installed mounts (RM, NF, FA, RF, TB) support the load. No back-up is required.

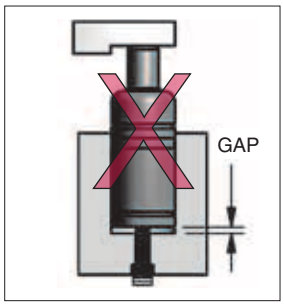


Retain inverted cylinders as shown with M6 cap screw. A close tolerance hole is required, depth > C/2.

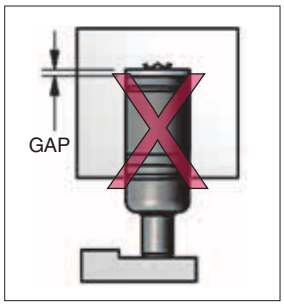


Mounts such as the DADCO-LOK may be used to retain the spring from rod end. If possible use a positive stop.

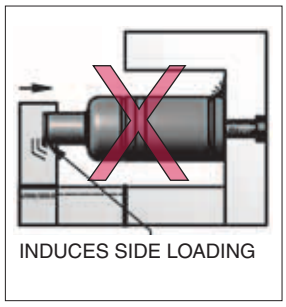
Improper Installation Examples



Verify the cap screw length.

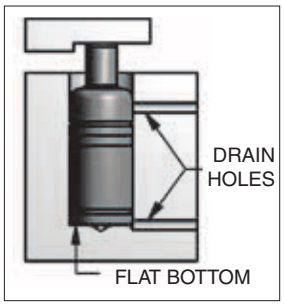


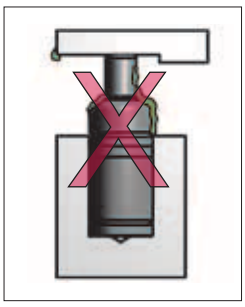
Avoid large gaps in the upper. Use the tapped hole in the base to secure and preload if possible.



Do not constrain the rod end. Do not use the bottom mount in an unsupported or open mounting application.

Contaminants

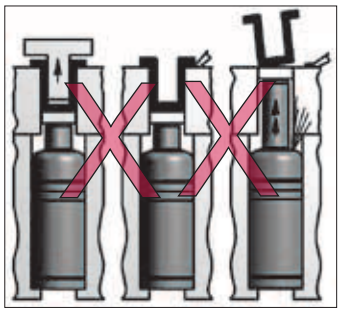




Provide adequate drainage in gas spring pockets. Direct contact with certain die lubricants and cleaners can be harmful to gas springs or may cause pressure increase.

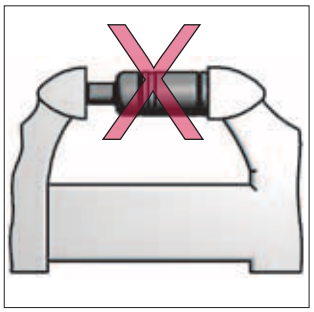
Uncontrolled Release

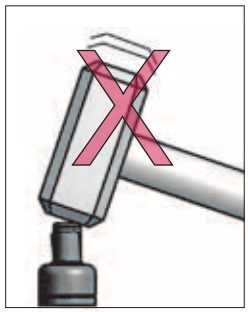
If parts are jamming, determine the root cause and repair it before production continues. Failure to repair the problem will cause failure or damage of the gas spring. Preloading the pad will prevent gas spring damage from “snap action” or sudden release. Restricting rod travel will help prevent spring damage.





Sudden release will cause the gas spring to exhaust.





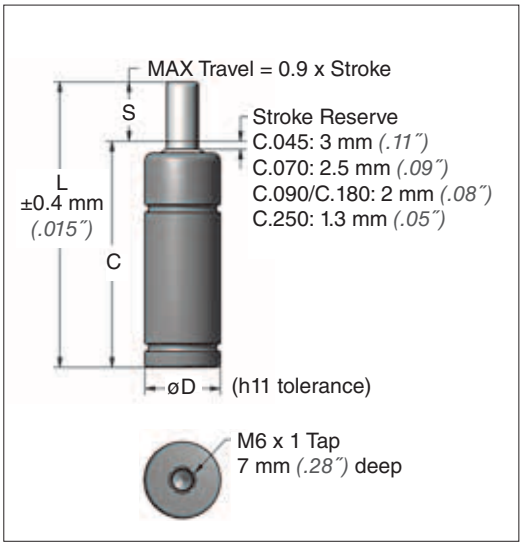
Never compress the gas spring in a vice or clamp outside of the die. Never strike the rod with a hammer to test for pressure; damage can result.

Technical Data

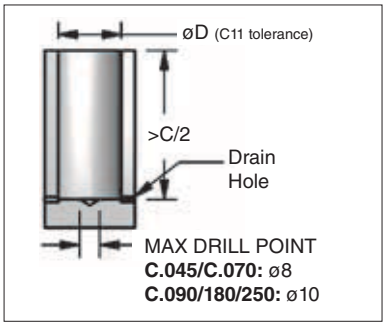
Operating Specifications

Maximum Charging Pressure		Charging Medium:	Nitrogen Gas
E.16 and E.24:	150 bar *(2175 psi)*	Operating Temperature:	-6°C – 71°C *(20°F – 160°F)*
Micro 45® – **Micro 250®**:	177 bar *(2560 psi)*		
SL.16:	180 bar *(2600 psi)*	Maximum Speed:	35 m/min *(23 in/sec)*

General Information



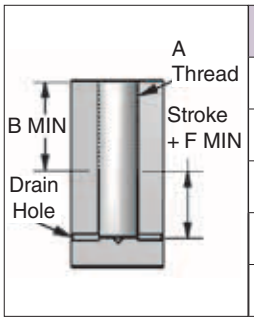
- DO NOT exceed 90% of stroke
- Stripping applications require a slight preload 0.5 mm – 1 mm
- Use enough force to strip the part
- Design adequate safety so spring is not over stroked



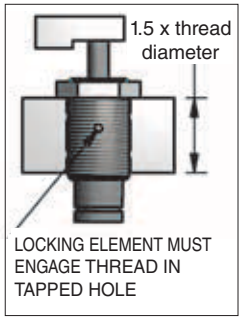
Stroke (mm)	SPM Limit
7-16	200
25-38	120
50-63	80
> 80	50

Travel 90% of nominal stroke

Threaded Body Installation Recommendations



Model	A	B	F	Model	A	B	F
E.16	M16 x 1.5	24 *.94*	12 *.47*	C.045.__.TB3	M16 x 2	35 *1.38*	5 *.20*
E.24	M24 x 1.5	35 *1.38*	25 *.98*	C.045.__.TB4	M16 x 2	24 *.94*	5 *.20*
SL.16	M16 x 1.5	24 *.94*	20 *.79*	C.090.__.TB1	1″-8	38 *1.50*	13 *.51*
C.045.__.TB1	5/8″-11	24 *.94*	5 *.20*	C.090.__.TB2	M24 x 1.5	36 *1.42*	13 *.51*
C.045.__.TB2	M16 x 1.5	24 *.94*	5 *.20*	C.090.__.TB3	M24 x 1.5	35 *1.38*	13 *.51*



A minimum thread engagement of 1.5 x thread diameter is recommended for threaded body cylinders.

Model	Maximum Installation Torque*
E.16	500 lb-in (56 N-m)
E.24	500 lb-in (56 N-m)
SL.16	500 lb-in (56 N-m)
C.045.__.TB1	200 lb-in (23 N-m)
C.045.__.TB2	500 lb-in (56 N-m)
C.045.__.TB3	300 lb-in (34 N-m)
C.045.__.TB4	300 lb-in (34 N-m)
C.090.__.TB1	500 lb-in (56 N-m)
C.090.__.TB2	500 lb-in (56 N-m)
C.090.__.TB3	500 lb-in (56 N-m)

**Based on strength of threads.*

DADCO®

43850 Plymouth Oaks Blvd. • Plymouth, MI • 48170 • USA
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